ELECTRONICS AND COMMUNICATION ENGINEERING

COMMUNICATION SYSTEM



Baseband Transmission

Lecture No.: 1

By- SAKET VERMA SIR



1. Random Process
2. Practice Questions



1. Block Diagram of Communication
2. Sampling Theorem

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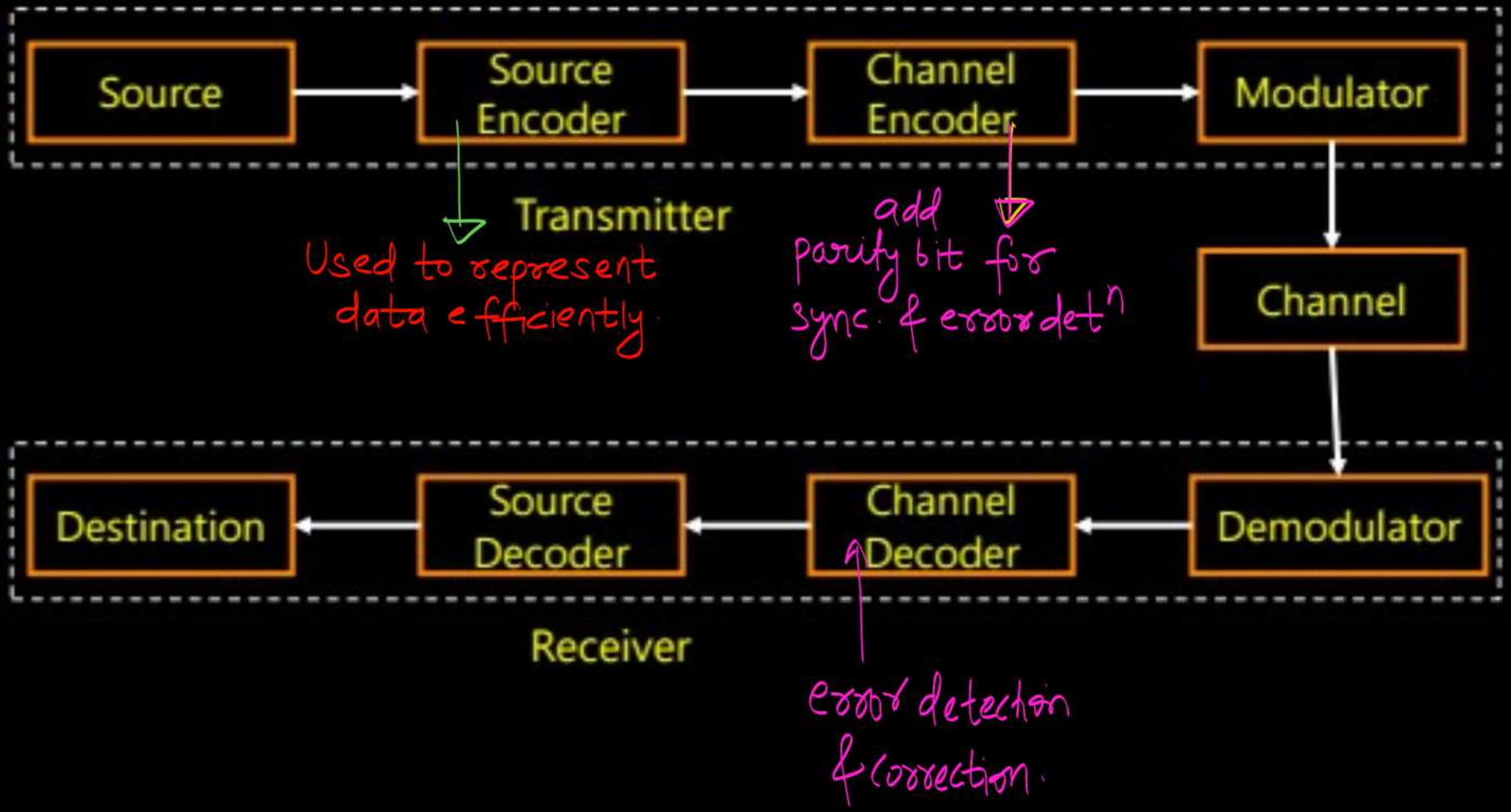




Necessity of Digital Communication

- The conventional methods of communication uses analog signals for long distance communications, which suffer from many losses such as distortion, interference, and other losses including security breach.
- In order to overcome these problems, the signals are digitized using different techniques. The digitized signals allow the communication to be clearer and more accurate without losses.

Block Diagram of Digital Communication





Block Diagram of Digital Communication

1. Source: A source can be any object or device that generates or holds the information Signal.
2. Source Encoder: The source encoder compresses the data into minimum number of bits. This process helps in effective utilization of the bandwidth and provide an efficient representation of data.
3. Channel Encoder: During the transmission of the signal, due to the noise in the channel, the signal may get altered and hence to avoid this, the channel encoder adds some redundant bits to the transmitted data. These are the error correcting bits.
4. Modulator: Modulator converts the signal from low – pass to High – pass in order to make it travel through the channel or medium.



Block Diagram of Digital Communication

5. Channel: The channel or a medium, allows the signal to transmit from the transmitter end to the receiver end.
6. Demodulator: Demodulator converts the signal from High – pass to Low – pass and reconstructed.
7. Channel Decoder: It detects and correct the errors which might occur during the transmission.
8. Source Decoder: The resultant signal is once again digitized by sampling and quantizing so that the pure digital output is obtained without the loss of information.
9. Destination: Here, the output is produced after the whole process.



Classification of Digital Communication

- Digital Communication
  - Baseband Transmission
    - **Analog**
      - PAM
      - PWM
      - PPM
    - **Digital**
      - PCM ✓
      - DPCM ✓
      - DM ✓
      - ADM
  - Bandpass Transmission
    - **Digital**
      - Amplitude Shift Keying (ASK)
      - Phase Shift Keying (PSK)
      - Frequency Shift Keying (FSK)
      - M - ary Shift Keying

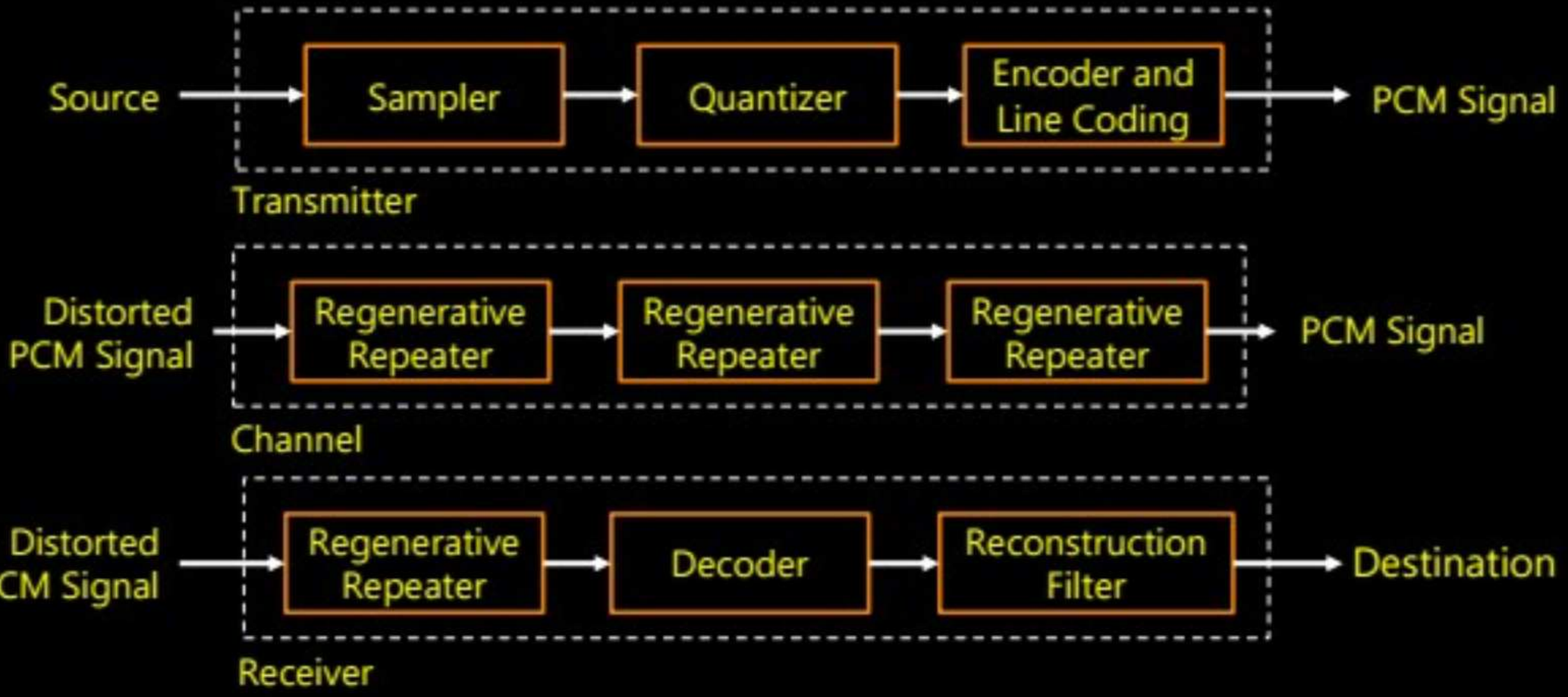


Baseband Transmission

- Baseband transmission is a transmission of a message data stream without modulating it with a high frequency carrier.
- Baseband transmission is a transmission of the encoded signal using its own baseband frequencies. It is used for short distance communication.



Block Diagram of Pulse Code Modulation





Advantages of Digital Communication

- Reliable communication.
- The correction and detection of errors are easy in digital communication, as there is a use of channel coding.
- As compared to analog signals, it is easy to save and retrieve digital signals.
- Digital communication is cheaper and simpler compared to analog signals because of the advancement of IC technologies.
- Multiplexing is efficient.
- To maintain the secrecy of information, the signal processing functions like compression and encryption are employed in digital circuits.

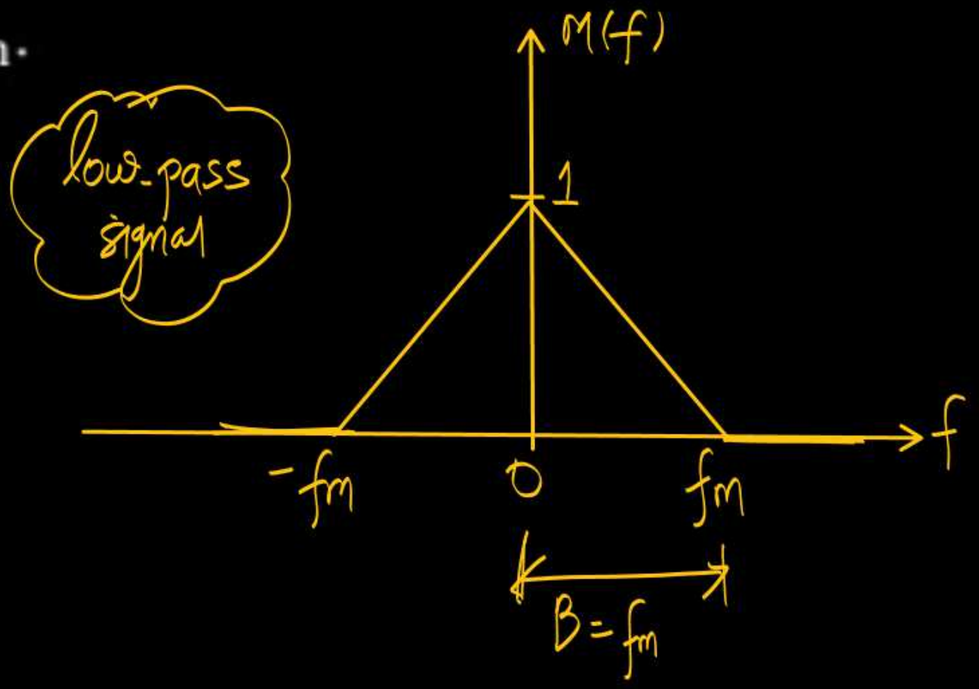


Disadvantages of Digital Communication

- There is high power consumption in digital communication.
- The most common limitation of digital communication is that it requires more transmission bandwidth.
- There is a requirement for synchronization in digital communication.
- Quantization Error

Sampling Theorem

- Sampling theorem states that a signal bandlimited to f_m Hz i.e., having no frequency component higher than f_m Hz, can be completely reconstructed by its samples if samples are taken at the interval of $T_s \leq \frac{1}{2f_m}$ sec or at the rate of $f_s \geq 2f_m$ $2f_m$.



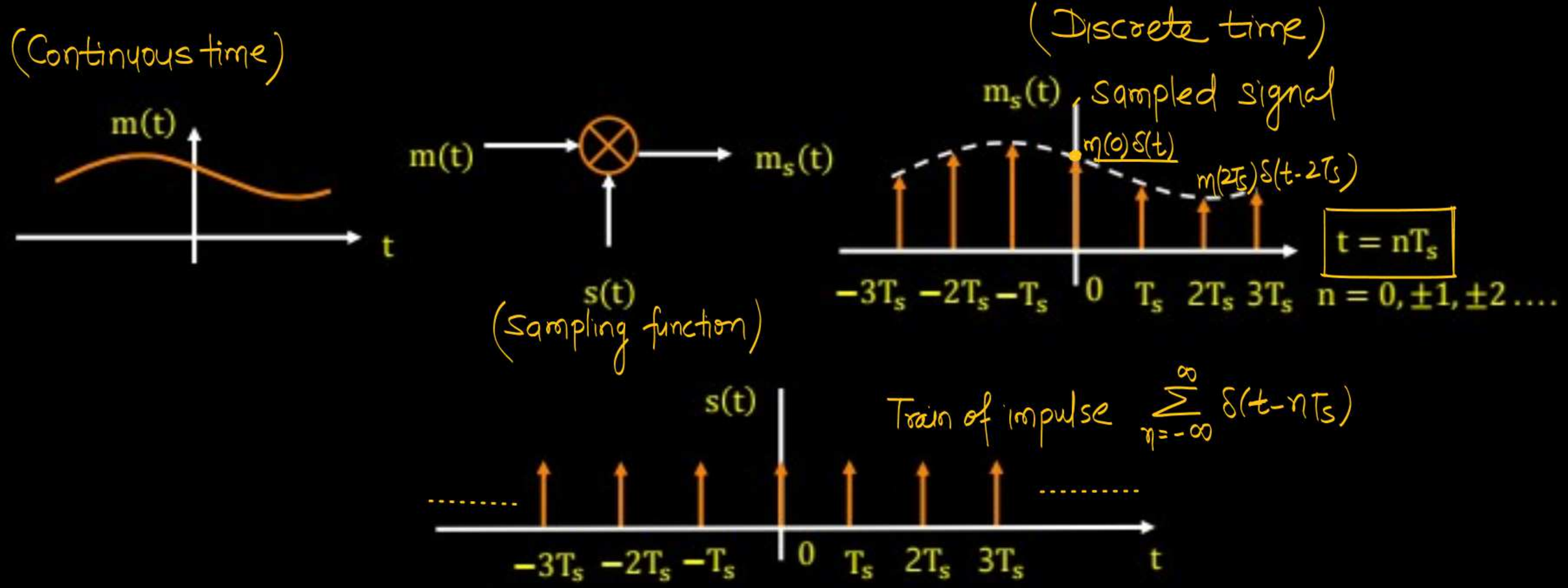
signal is bandlimited to fm.

signal has Bandwidth fm.

signal has highest freq fm.

Sampling Theorem

- Consider a signal $m(t)$ passed through a system shown below :-



$$m_s(t) = m(t) \cdot s(t)$$

$$= m(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$\boxed{m_s(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\,\delta(t - nT_s)}$$

$n=0, \quad m_s(n) = m(0) \cdot \delta(t).$

$s(t)$, is a periodic signal with Time period, T_s.

$$s(t) = \frac{1}{T_s} + \frac{2}{T_s}\cos\omega_s t + \frac{2}{T_s}\cos 2\omega_s t + \cdots\cdots\cdots\cdots$$

$$m_s(t) = m(t)\,s(t) = \frac{m(t)\,1}{T_s} + \frac{2}{T_s}m(t)\cos\omega_s t + \frac{2}{T_s}m(t)\cos 2\omega_s t + \cdots\cdots$$

Taking F.T.

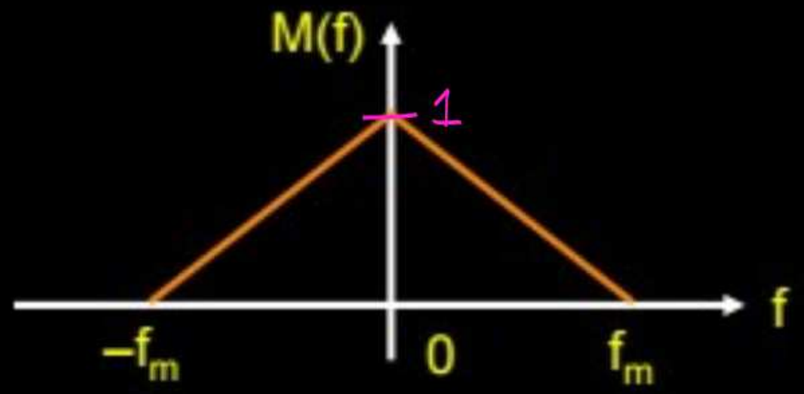
$$M_s(f) = \underbrace{\frac{M(f)}{T_s}}_{n=0} + \underbrace{\frac{M(f-f_s)}{T_s}}_{n=1} + \underbrace{\frac{M(f+f_s)}{T_s}}_{n=-1} + \underbrace{\frac{M(f-2f_s)}{T_s}}_{n=2} + \underbrace{\frac{M(f+2f_s)}{T_s}}_{n=-2} + \cdots$$

$$\boxed{M_s(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} M(f - nT_s)}$$



Sampling Theorem

- Consider a signal m(t) bandlimited to f_m having spectrum M(f) as shown below :-



Let f_s be the Sampling frequency, then there exist 3 different cases

Case 1 : $f_s > 2f_m$ (Oversampling)

Case 2 : $f_s = 2f_m$ (Critical sampling)

Case 3 : $f_s < 2f_m$ (Under sampling)

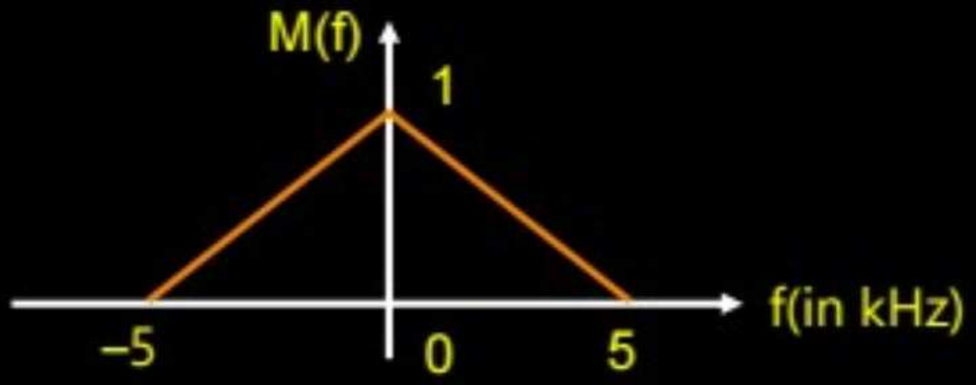
Sampling Theorem

$$M_s(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} M(f - nf_s)$$

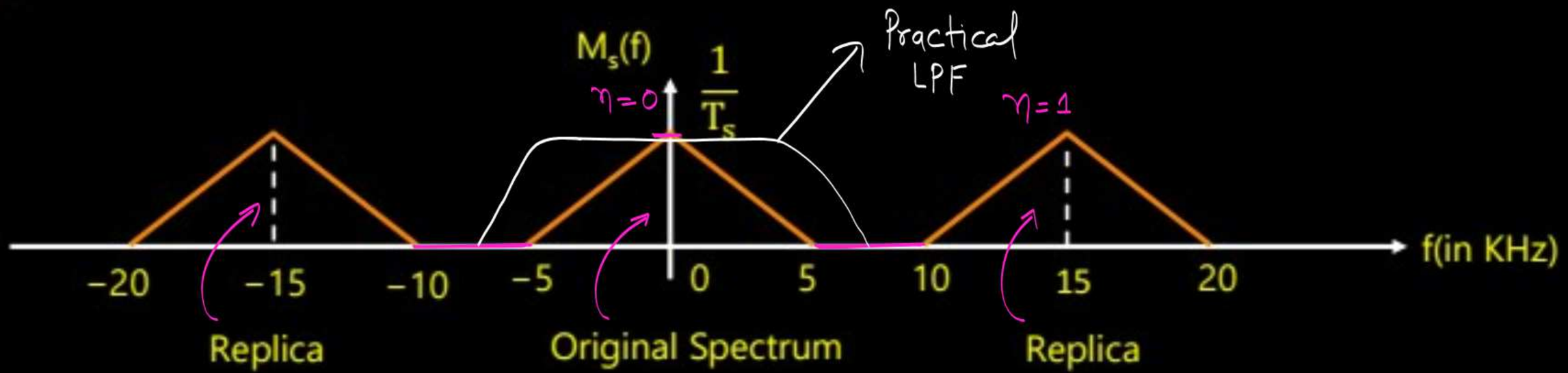
- Case 1: $f_s > 2f_m$ (Oversampling)

Consider a Signal bandlimited to 5 kHz as shown and Sampling frequency is 15 kHz

$f_m = 5\,kHz$

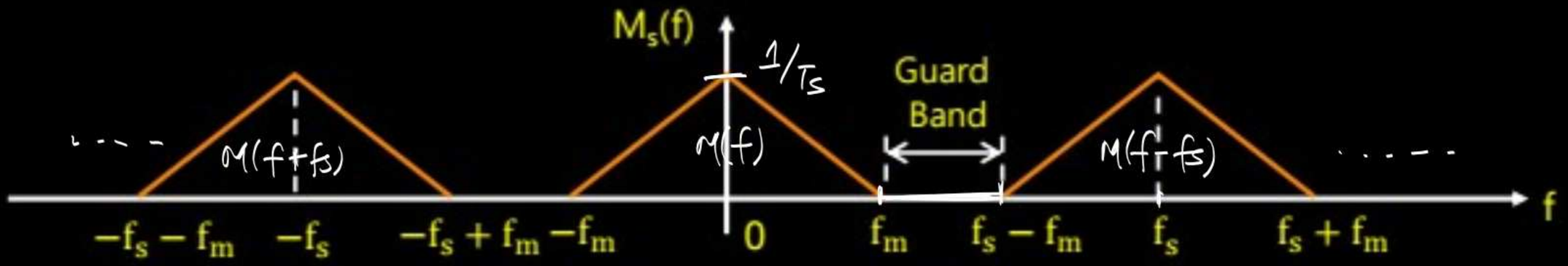


Sampling Theorem



Sampling Theorem

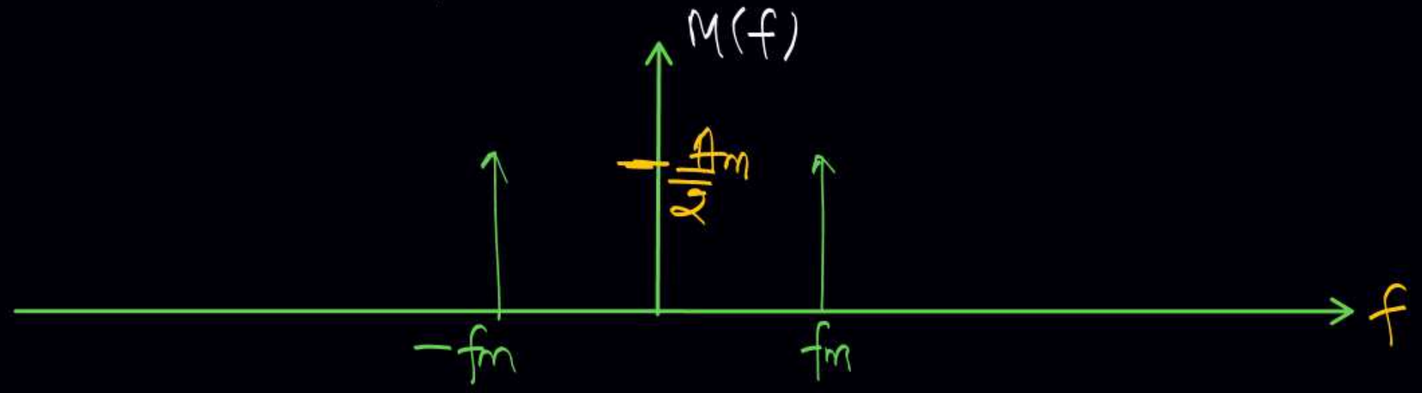
$f_s > 2f_m$

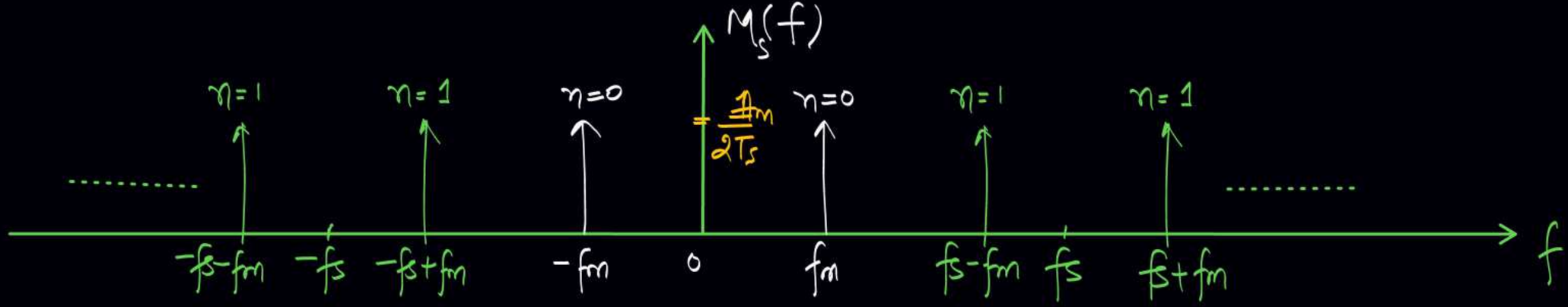


Guard Band $= f_s - f_m - f_m$

$GB = f_s - 2f_m$

For single tone, $m(t) = A_m \cos \omega_m t$





After sampling of single tone frequency components present in sampled signal are

$f_s,\ f_s \pm f_m,\ 2f_s \pm f_m,\ 3f_s \pm f_m$



Sampling Theorem

- Frequency components that are present in spectrum of sampled signal in case of single tone modulating signal for oversampling are :-

$$f_m , f_s \pm f_m , 2f_s \pm f_m , 3f_s \pm f_m , \ldots\ldots$$

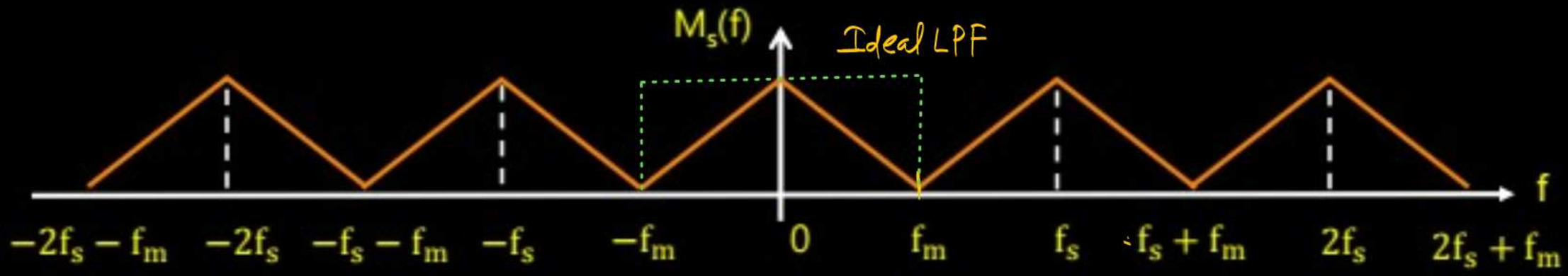
- In case of Oversampling, it is possible to recover the original message signal from the sampled signal using practical Lowpass Filter.

Sampling Theorem

- Case 2: $f_s = 2f_m$ (Critical sampling) ⟶ Nyquist Rate.

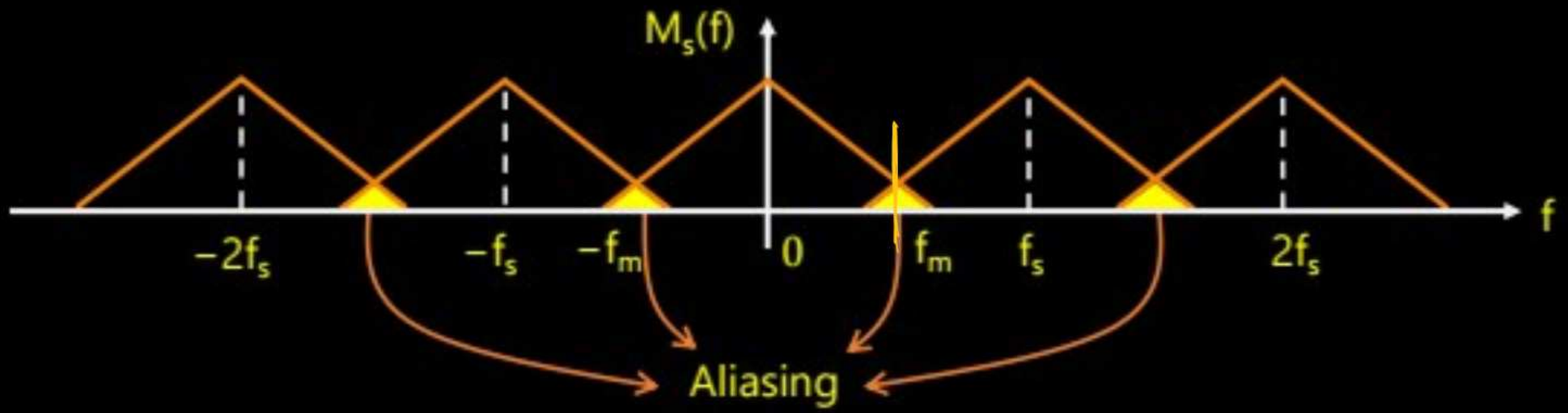
Consider a Signal bandlimited to 5 kHz as shown and Sampling frequency is 15 kHz

$$M_s(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} M(f - nf_s)$$



Sampling Theorem

- Case 3: $f_s < 2f_m$ (Under sampling)



- In case of Under sampling, it is not possible to recover the original message signal from the sampled signal due to aliasing effect.



Sampling Theorem

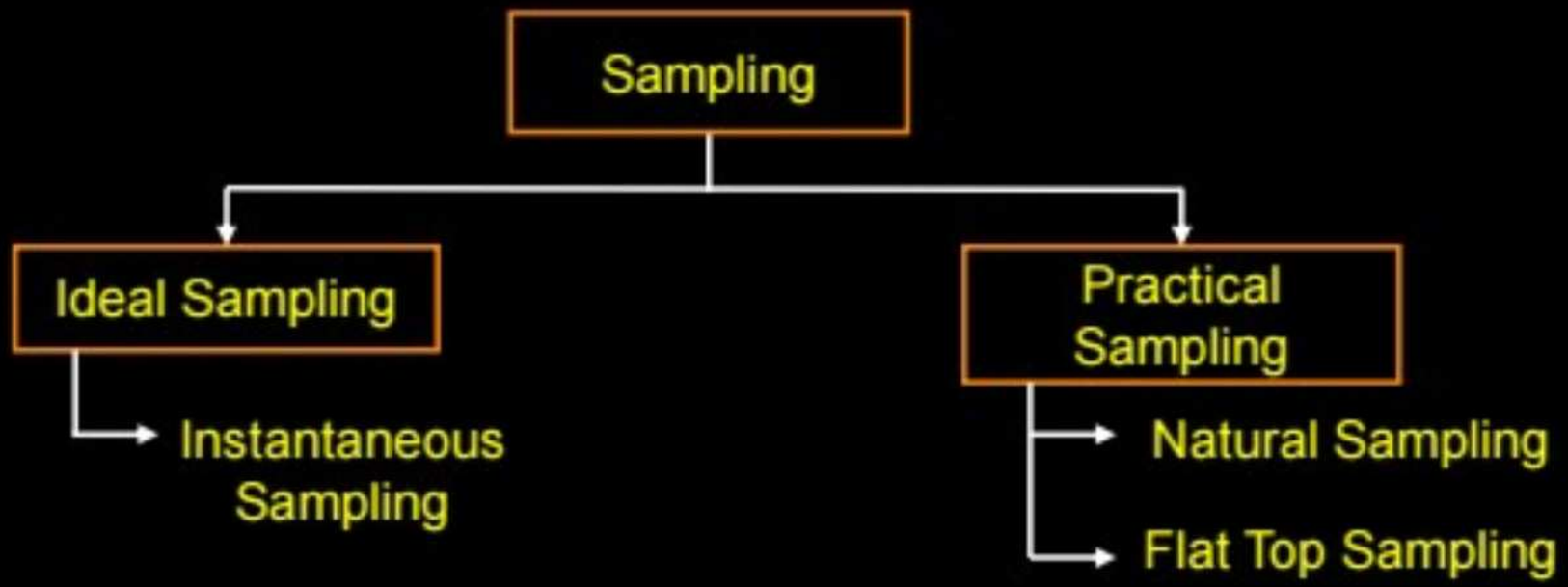
- Due to under sampling high frequency component of message signal gets distorted and this effect is known as Aliasing effect or Fold over error.

$$\text{Folding Frequency} = \frac{f_s}{2}$$

- Folding frequency is the frequency at which the spectrum folds into itself.



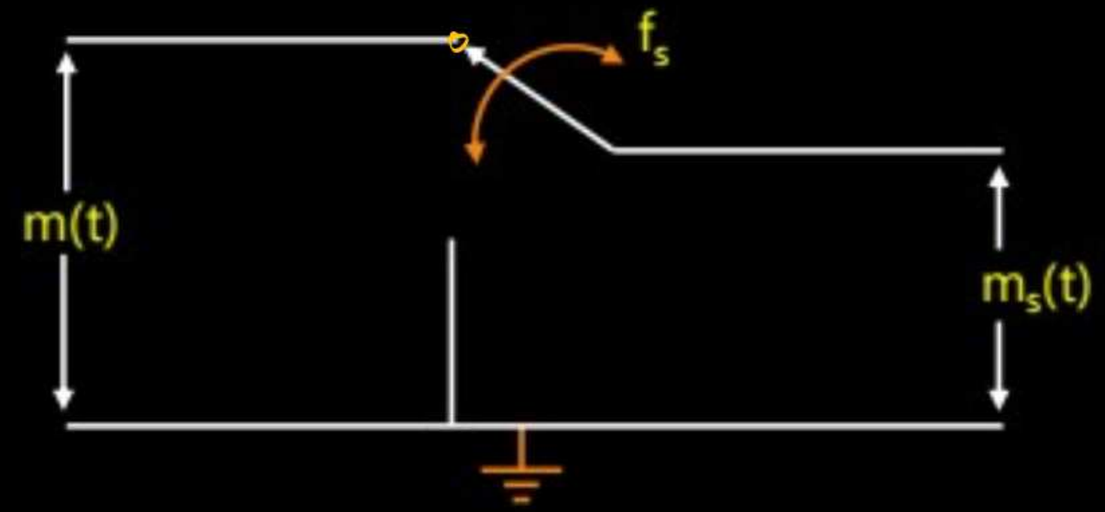
Types of Sampling





Instantaneous Sampling

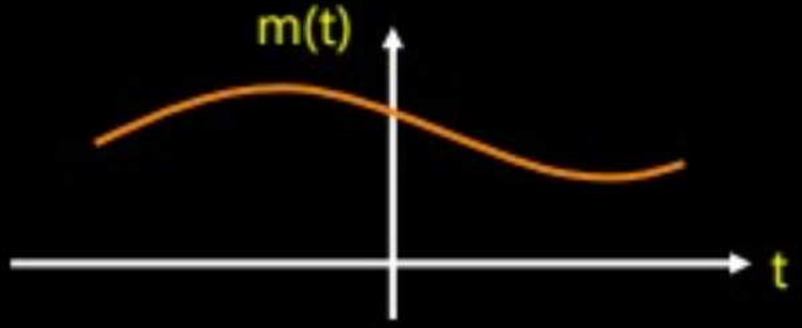
- In this type of Sampling, the sampling function is a train of Impulses having pulse width $\tau = 0$.
- Instantaneous sampling can be done by using a Switching Sampler Circuit as shown below :-

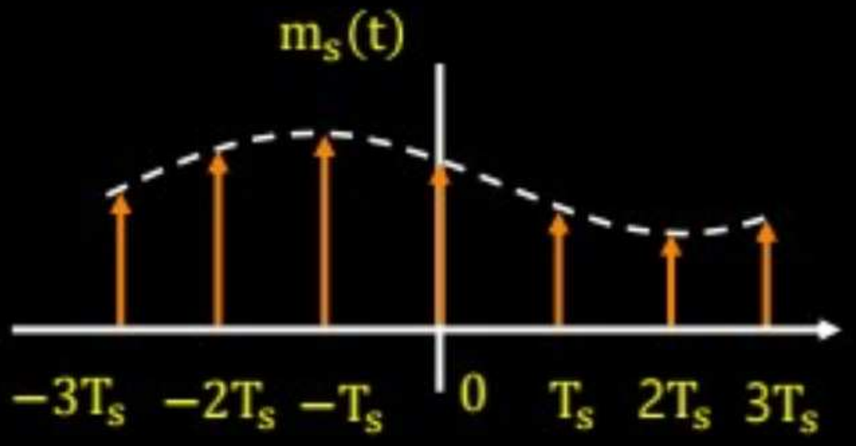


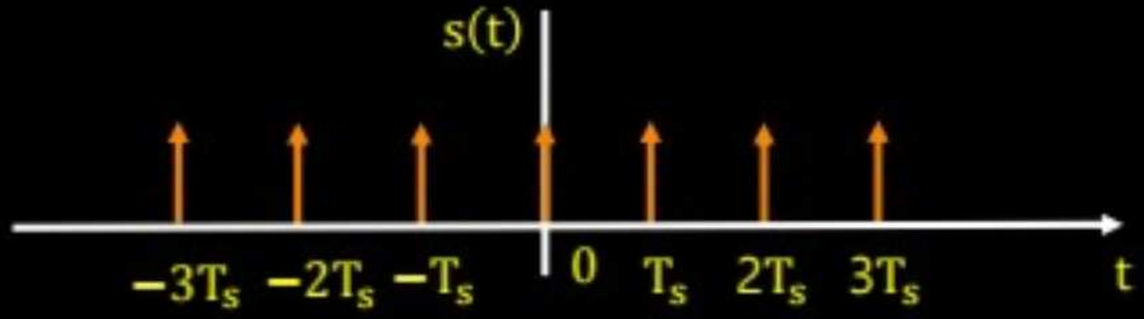
Switching Sampler

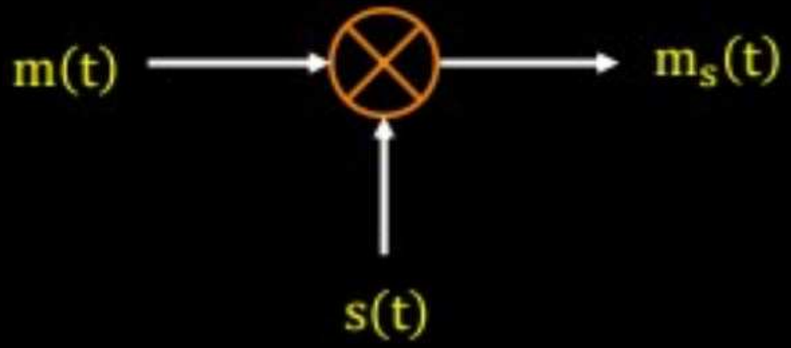


Instantaneous Sampling











Instantaneous Sampling

- The train of impulses s(t) can be expressed as

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- The Sampled signal $m_s(t)$ can be expressed as

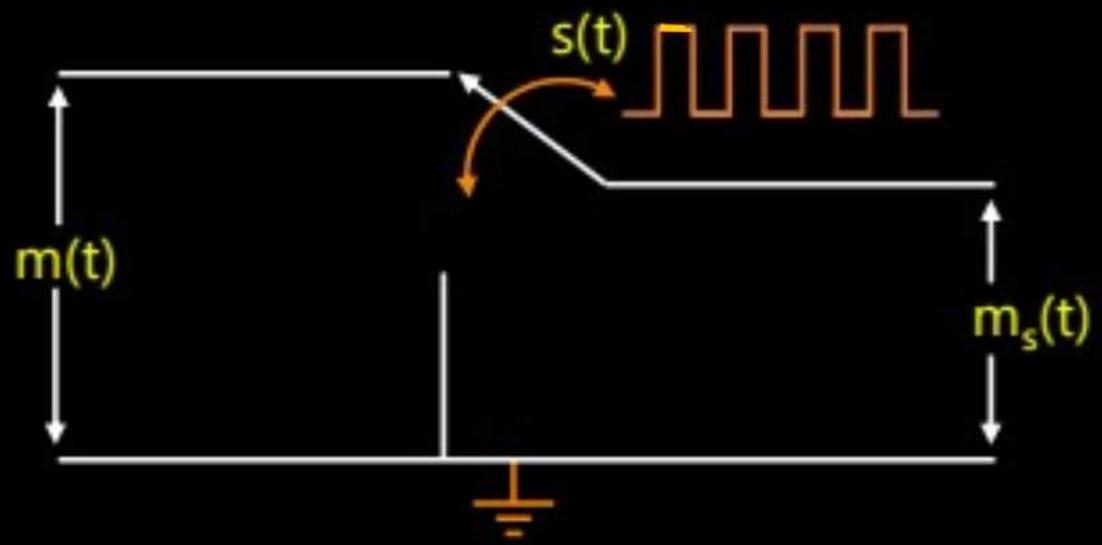
$$m_s(t) = m(t).s(t)$$

NOTE

- The instantaneous sampling has a train of impulses. The pulse width of the samples has almost zero value. Therefore, it has negligible power content and thus may not be used for transmission purpose.



Natural Sampling

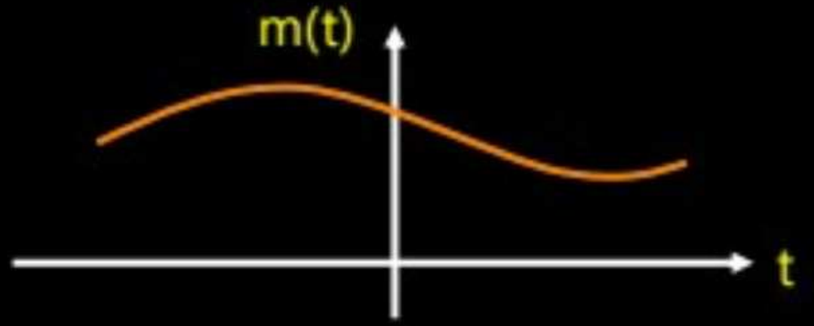


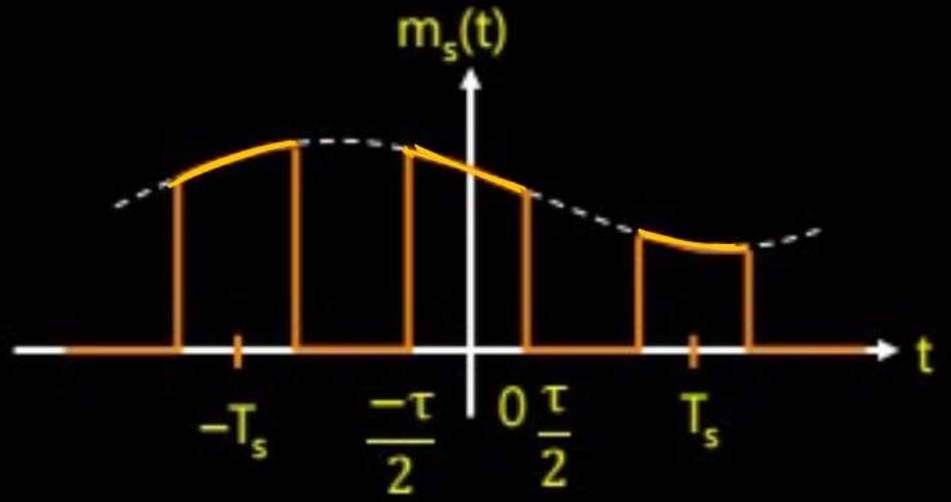
Natural sampling is a practical method. In this type of sampling, the pulse has a finite width equal to τ and Amplitude A.

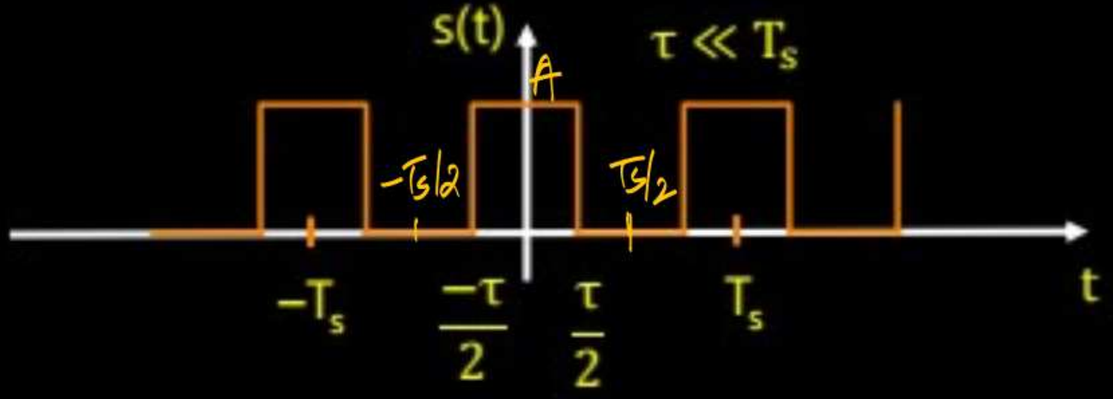
- When s(t) goes HIGH, Switch is closed
- $\therefore$ when $s(t) = A$, $m_s(t) = m(t)$
- $\therefore$ when $s(t) = 0$, $m_s(t) = 0$

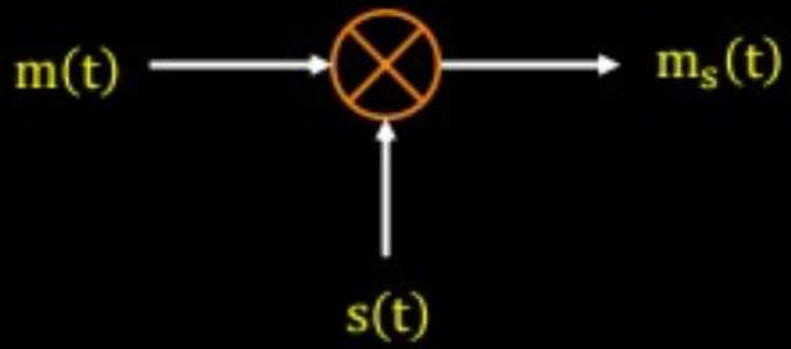


Natural Sampling









$s(t)$ is a periodic signal with Time period T_s.

$$s(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_s t}$$

$$c_n = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} s(t) e^{-jn\omega_s t} dt = \frac{1}{T_s} \cdot A \int_{-\tau/2}^{\tau/2} e^{-jn\omega_s t} dt$$

$$= \frac{A}{T_s} \cdot \frac{1}{n\omega_s} \left[\frac{e^{jn\omega_s \frac{\tau}{2}} - e^{-jn\omega_s \frac{\tau}{2}}}{j} \right]$$

$$= \frac{A}{n\pi} \sin\left(\frac{n\omega_s \tau}{2}\right) = \frac{A f_s \tau}{n\pi f_s \tau} \sin\left(n\pi f_s \tau\right) = \frac{A\tau}{T_s} \operatorname{sinc}(n f_s \tau)$$

$$m_s(t) = m(t) s(t) = \sum_{n=-\infty}^{\infty} \frac{A\tau}{T_s} \operatorname{sinc}(n f_s \tau) \cdot e^{jn\omega_s t} \, m(t)$$

Taking FT,

$$M_s(f) = \frac{A\tau}{T_s} \sum_{n=-\infty}^{\infty} \text{sinc}(n f_s \tau) \cdot M(f - n f_s)$$

$n = 0$, $M_s(f) = \frac{A\tau}{T_s} M(f)$

$n = 1$, $M_s(f) = \frac{A\tau}{T_s} \text{sinc}(f_s \tau) \cdot M(f)$

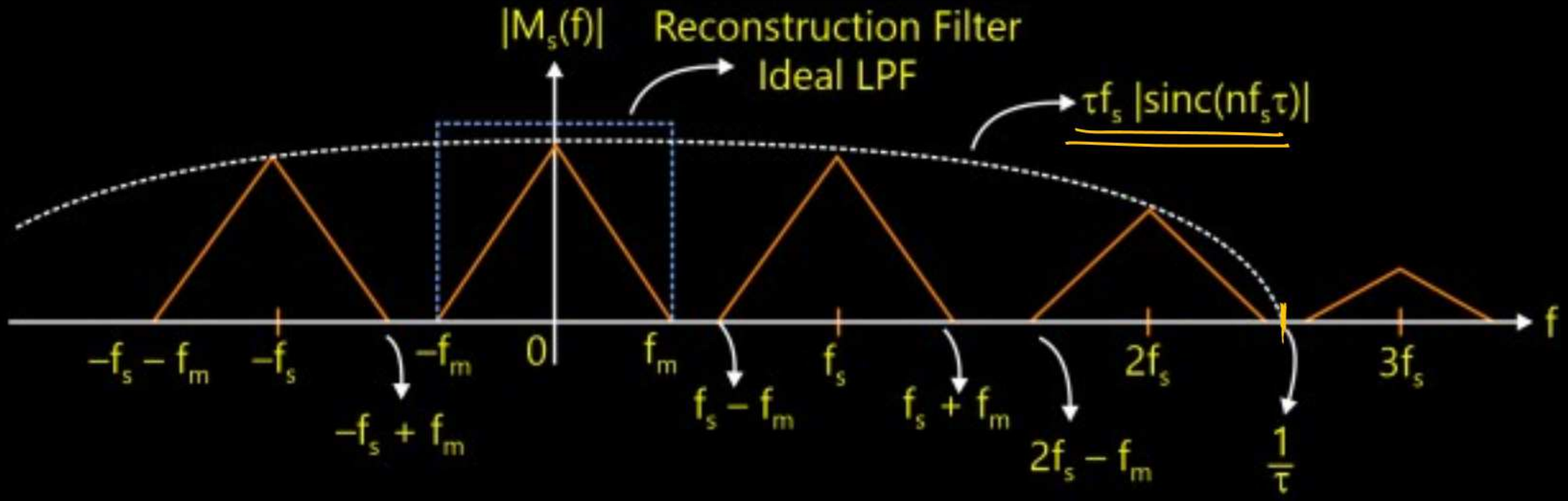
$\frac{\sin(\pi f_s \tau)}{\pi f_s \tau}$ = zero crossing

$\pi f_s \tau = n\pi$

$f_s = \frac{n}{\tau}$

$\tau = n T_s$

Natural Sampling



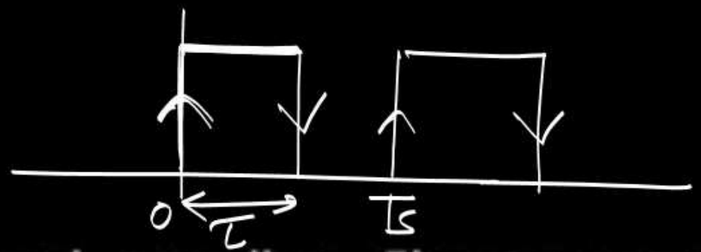


Natural Sampling

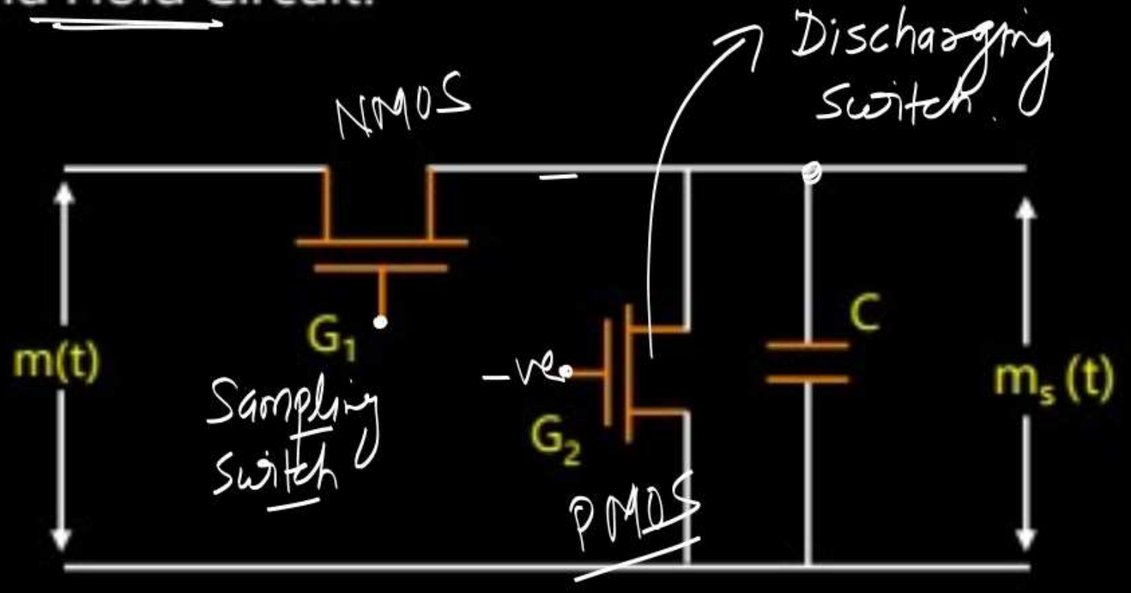
NOTE

- The main disadvantage of using natural sampling is that if the Noise interfere at the top of a transmission pulse, then on recovering the message signal with practical LPF, Noise also appears in the output of LPF along with recovered message signal.

Flat Top Sampling



- Flat top sampling is basically Superior than Natural sampling. Flat top sampling is based on Sample and Hold Circuit.



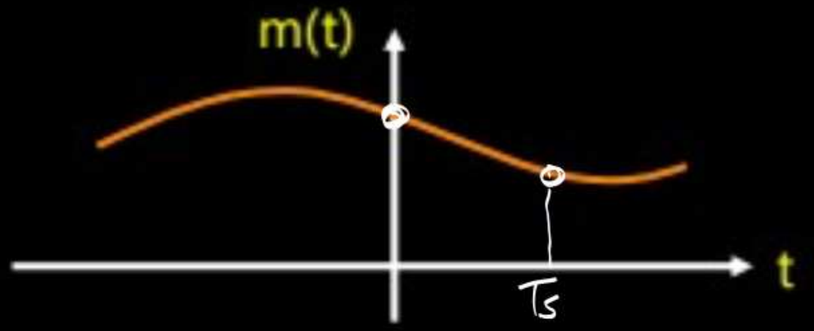
- CMOS is used as Switch. One is PMOS and other is NMOS.
- G_1 is known as Sampling Switch and G_2 is known as Discharging switch.

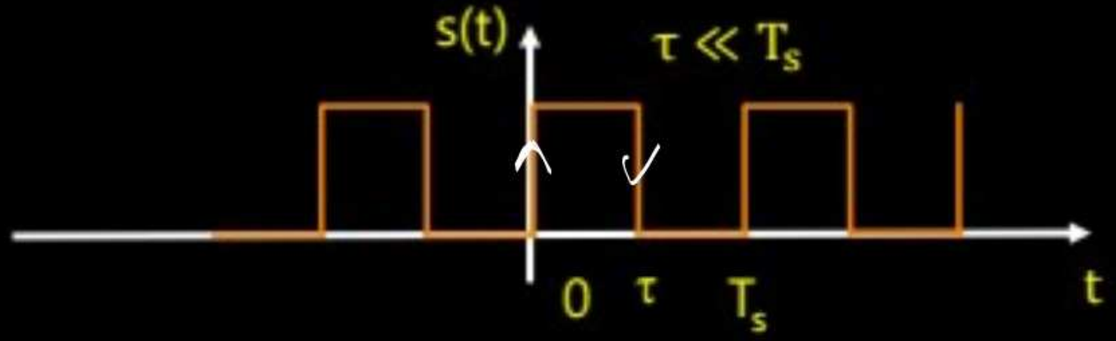


Flat Top Sampling

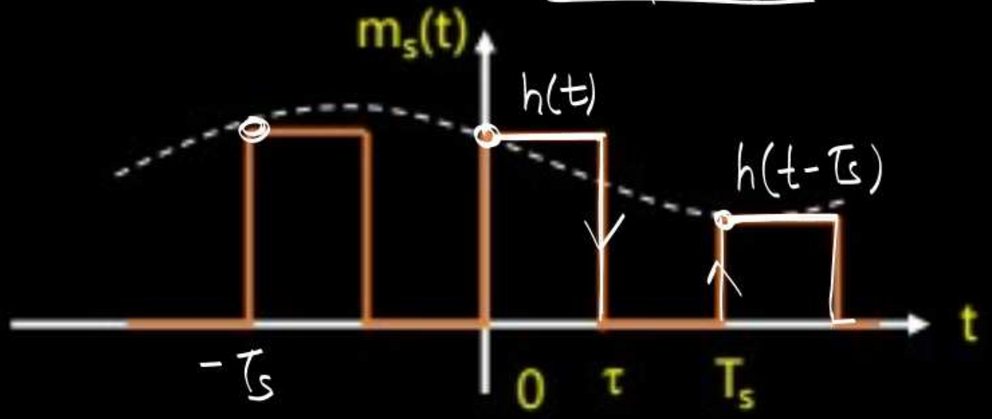
- A rectangular pulse is applied to G_1 at instant of sampling for very short duration.
- Both switch will remain open for duration of τ, and capacitor holds the instantaneous sampled value.
- At the end of τ interval, pulse is applied to G_2 and capacitor discharges through gate G_2.

Flat Top Sampling





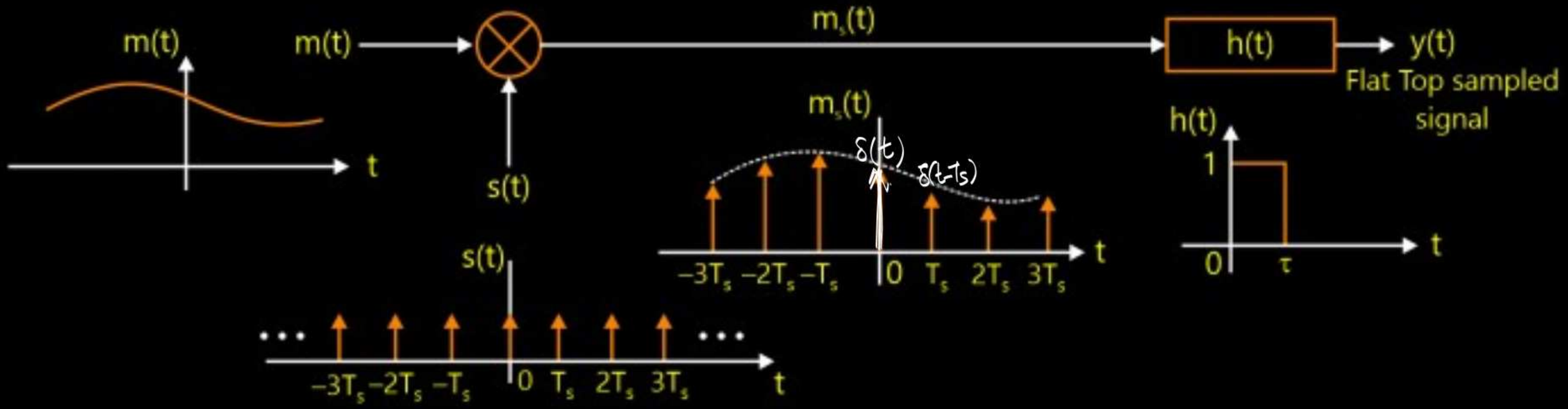
flat top signal.



Flat Top Sampling

$h(t) * \delta(t) = h(t)$

$h(t) * \delta(t - T_s) = h(t - T_s)$



$$m_s(t) = m(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$y(t) = m_s(t) * h(t)$$

$$= m(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) * h(t)$$

$$= \sum_{n=-\infty}^{\infty} m(nT_s)\, \delta(t - nT_s) * h(t)$$

$$y(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\, h(t - nT_s)$$

$n = 0,\quad y(t) = m(0)\, h(t)$

$n = 1,\quad y(t) = m(T_s)\, h(t - T_s)$

$\vdots$

FT

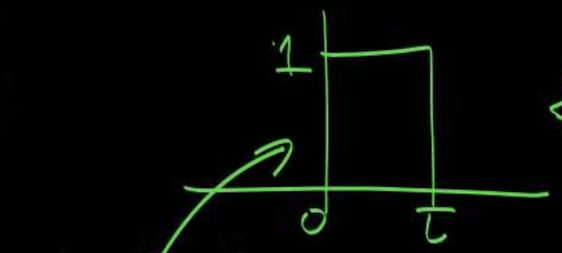
$$Y(f) = M_s(f)\, H(f)$$

$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} M(f - nf_s) \cdot H(f)$$

$n = 0,$

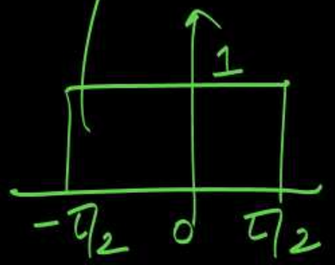
$$Y(f) = \frac{M(f) \cdot H(f)}{T_s}$$

Flat Top Sampling



$\xleftrightarrow{FT} e^{-j2\pi f\frac{\tau}{2}}\,\tau\,\text{sinc}(f\tau)$

- Consider a signal bandlimited to f_m having spectrum as shown below and the sampling frequency $f_s \gg 2f_m$

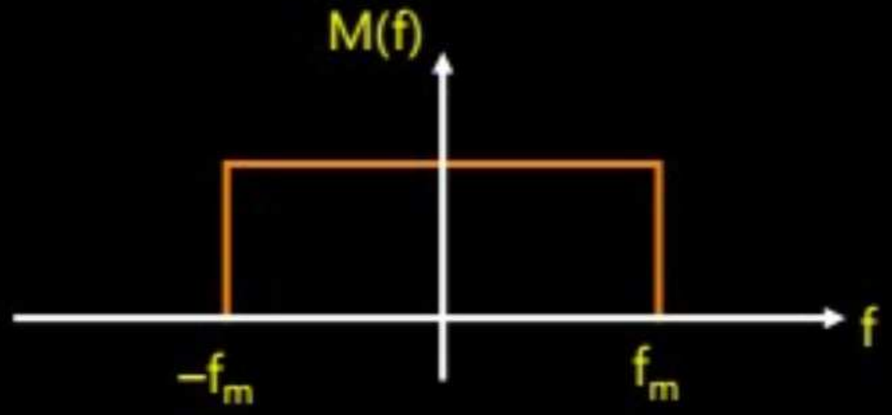


$|H(f)| = \tau\,\text{sinc}(f\tau)$

Zero cross.

$\pi f\tau = n\pi$

$$f = \frac{n}{\tau}$$ n = integer

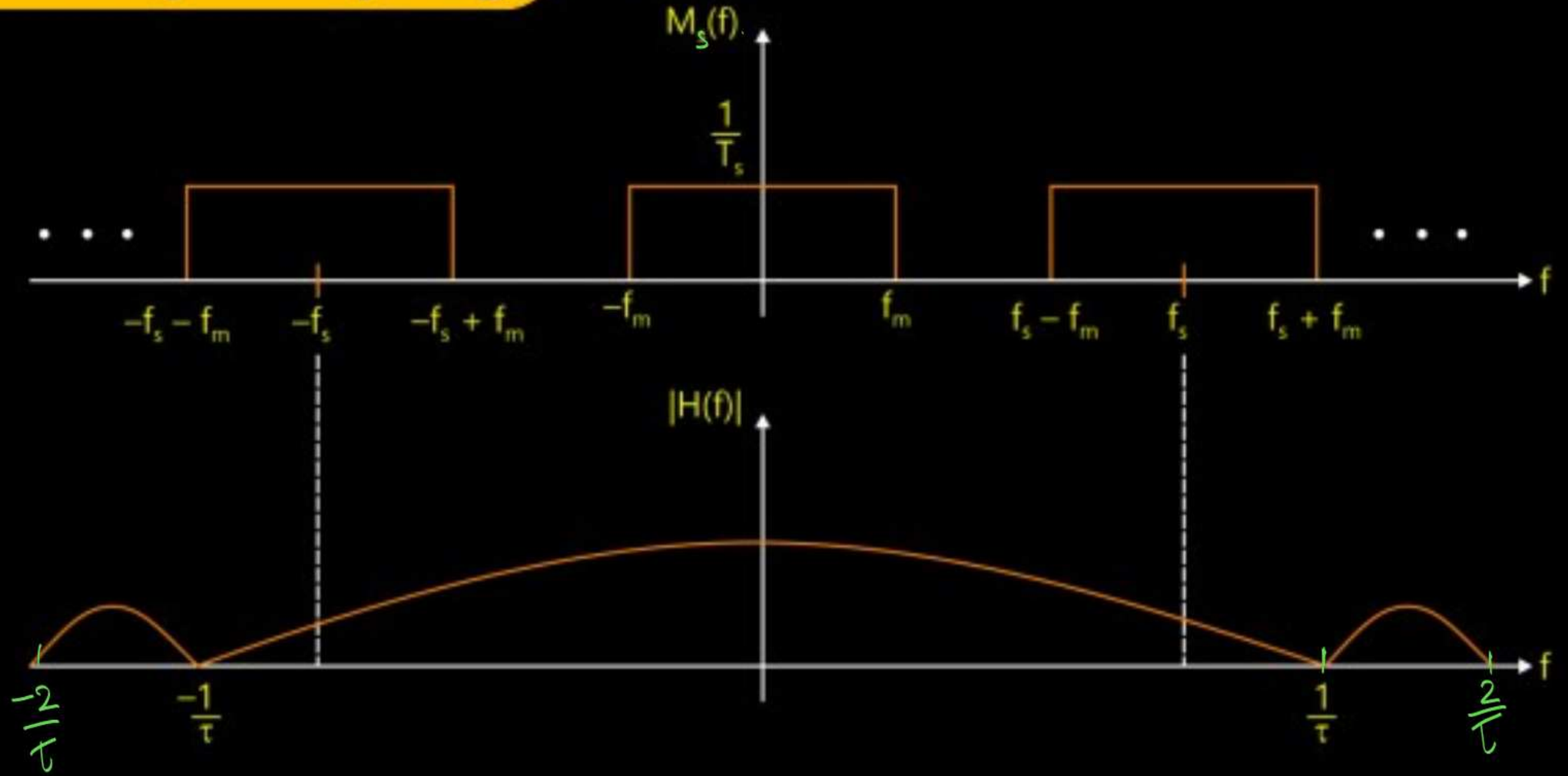


$\tau \ll T_s$

$\tau \ll \frac{1}{f_s}$

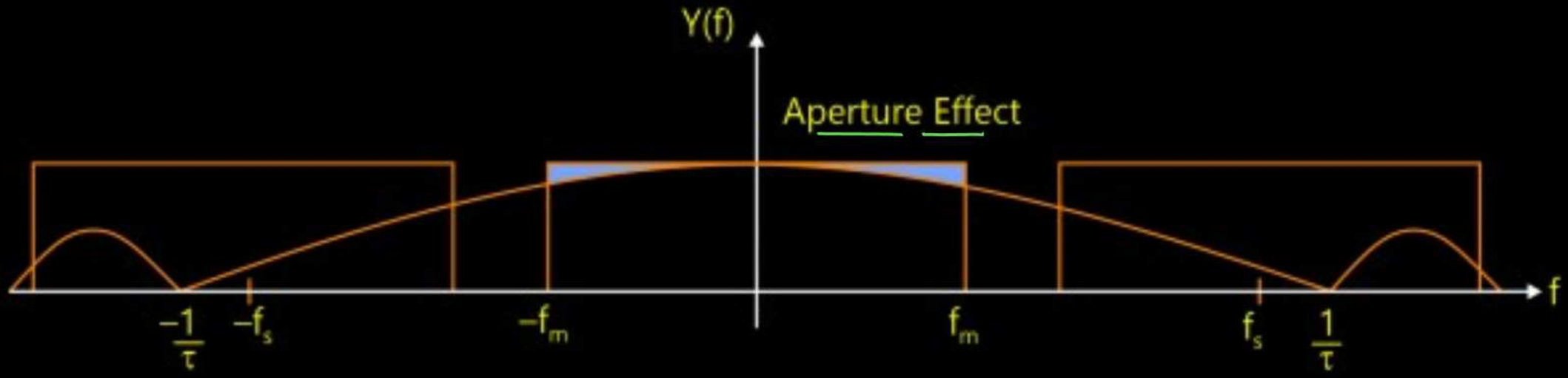
$$f_s \ll \frac{1}{\tau}$$

Flat Top Sampling





Flat Top Sampling



- High frequency component of message signal gets distorted due to H(f), this effect is known as Aperture effect.

To Avoid aperture effect $\frac{\tau}{T_s} \ll 0.1$

Flat Top Sampling

- To recover message signal, we use Equalizer Circuit.



Reconstruction Filter

$$|H_{eq}(f)| = \frac{1}{|H(f)|} = \frac{1}{|\tau\, \text{sinc}(f\tau)|}$$

THANK you